

Lab 03

Due Feb 7 5:00 PM PST

AUTHOR

Peter Khrissanov

Problem 1: Hazardous waste

(2 points)

```
## write answer here
library(RSQLite)
data <- read.csv("incinerated.new")
db <- dbConnect(SQLite(), dbname = "combined.sqlite")
dbWriteTable(
  conn = db,
  name = "incin",
  value = data[1:4, ],
  row.names = FALSE,
  overwrite = TRUE
)

head(dbReadTable(db, "incin"))
```

	Country	Year	Value
1	Armenia	2015	3767
2	Armenia	2014	2379
3	Armenia	2013	17
4	Austria	2014	215076

Part b)

(2 points)

```
## write answer here
data_insert <- tail(data, n = 3)

for(i in 5:7){
  query <- paste0("INSERT INTO incin VALUES ('",
    data$Country[i], "'", data$Year[i],
    "'", data$Value[i],
    "'", ")")

  dbExecute(db, query)
}
```

```
incin_data <- dbReadTable(db, "incin")
tail(incin_data, 3)
```

	Country	Year	Value
5	Azerbaijan	2015	210858.0
6	Azerbaijan	2014	111309.5
7	Azerbaijan	2013	86426.5

Part c)

(4 points)

```
gen <- read.csv("generated.new")
inc <- read.csv("incinerated.new")
land <- read.csv("landfilled.new")
rec <- read.csv("recycled.new")

for (i in seq_len(nrow(gen))) {
  query <- paste0(
    "INSERT INTO lab03combined (Country, Year, Generated, Incinerated, Landfilled, Recycled) VALUES (",
    gen$Country[i], ", ",
    gen$Year[i], ", ",
    gen$Value[i], ", ",
    inc$Value[i], ", ",
    land$Value[i], ", ",
    rec$Value[i],
    ")"
  )
  dbExecute(db, query)
}
```

Part d)

(2 points)

```
combined_df <- dbReadTable(db, "lab03combined")
tail(combined_df, 7)
```

	Country	Year	Generated	Incinerated	Landfilled	Recycled
53	Armenia	2015	555075.0	3767.0	551018.0	290.0
54	Armenia	2014	576419.4	2379.0	571750.4	2290.0
55	Armenia	2013	579049.8	17.0	574973.8	4059.0
56	Austria	2014	1272288.0	215076.0	60260.0	163911.0
57	Azerbaijan	2015	262556.5	210858.0	166965.1	5174.9

58	Azerbaijan	2014	456625.1	111309.5	414633.6	3163.4
59	Azerbaijan	2013	202882.0	86426.5	65401.1	618.4

Part e) (Optional Bonus)
